

Guideline Title: Intracranial Pressure Invasive Monitoring

Guideline Number: GUID-3203-PR014

Issue date: 09/09/2014	Replaces Dept. Policy:
Revision dates: 4/30/24	Developed by: Air Care Team
Approved by: Dr. Christopher Hunter, MD Air Care Team Medical Director	Approved by:
Signature: 	Signature:
Department Numbers: 3203	

- I. **PURPOSE:** **PURPOSE:** Continual monitoring of the intra-cerebral pressure in the cranial vault.
- II. **DEFINITIONS:** ICP – Intracranial Pressure
- III. **DEPARTMENT GUIDELINE:**
 - a. Assess Patient: Keep neck in neutral position. Keep head of bed at 30 degrees. Assess that dressing is clean, dry, and intact.
 - b. Assess Device (Always use personal protection equipment/gloves during the procedure).
 - i. Assess all stopcocks to make sure nothing is clamped.
 - ii. Assess filter at the top of the collection device. It must not get wet. Do not lay device flat.
 - iii. Assess waveform – P1, P2, P3. Assess for kinks, air bubbles, blood clots, or debris.
 - iv. Evaluate the invasive monitor system from referring hospital to ensure that ventricular catheter is functioning correctly and accurately. Verify that a patent transducer system is in place. Use Preservative free Normal Saline only, do not add Heparin to the system.
 - c. Zero the device.
 - i. Place and secure with tape the transducer level with the outer aspect of the ear canal and the lateral canthus of the eye on the same side as the ventriculostomy catheter is inserted on. Connect the transducer to the invasive monitor using the invasive monitoring cable.
 - ii. Zero the ventriculostomy line. Turn the stopcock closest to the insertion site off to the patient and open to air. Observe the waveform on the monitor flatten. Press the zero button. The monitor will calibrate and “zeroed” will appear on the monitor screen.
 - iii. Replace the cap and open the stopcock to begin transducing the ICP. Once the ventriculostomy has been “zeroed” observe the monitor for a normal waveform.
 - d. Monitor the patient
 - i. Do not flush pressure lines while connected to patient.
 - ii. Close the drain for all patient movements and altitude changes.
 - iii. If the patient sustains an ICP > 15 cm/H₂O for more than 5 minutes consult the receiving physician.
 - iv. If the device is inadvertently pulled out, place a sterile dressing over the insertion site.
- IV. **DOCUMENTATION:** EMS Charts

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V. **REFERENCES:** None